



Cardiac Scan AI

Heart Disease Prediction Using Machine Learning: A
Comparative Analysis



A glowing heart with a blue grid background and molecular structures. The heart is rendered in a 3D style with a blue and white color scheme, featuring a bright blue glow. It is surrounded by a blue grid pattern and several molecular structures, including a prominent one in the center-right. A white line graph, resembling an ECG, is overlaid on the heart, extending from the left side towards the right. The overall background is a dark blue gradient with a subtle grid pattern.

Project Statement

Heart disease is a leading cause of death worldwide, and early detection is crucial for prevention. This project uses machine learning models to predict heart disease based on patient health data, enabling faster and more accurate diagnosis.

Objective:

- Develop an ML-based system to predict heart disease.
- Compare different ML models to find the most accurate one.
- Use patient health parameters (age, blood pressure, cholesterol, glucose, etc.) to make predictions.



Methodology



Data Collection

Gather relevant data from diverse sources.



Preprocessing

Clean, normalize, and engineer data.



Model Selection

Choose and train ML/DL algorithms for peak performance.



Model Development

We design and implement machine learning models tailored to the specific requirements of the project. This involves experimenting with different architectures, hyperparameters, and optimization techniques.



Logistic Regression

Simple, Interpretable,
Probabilistic, Linear



SVM

Accurate, Robust, Linear,
Margin-based



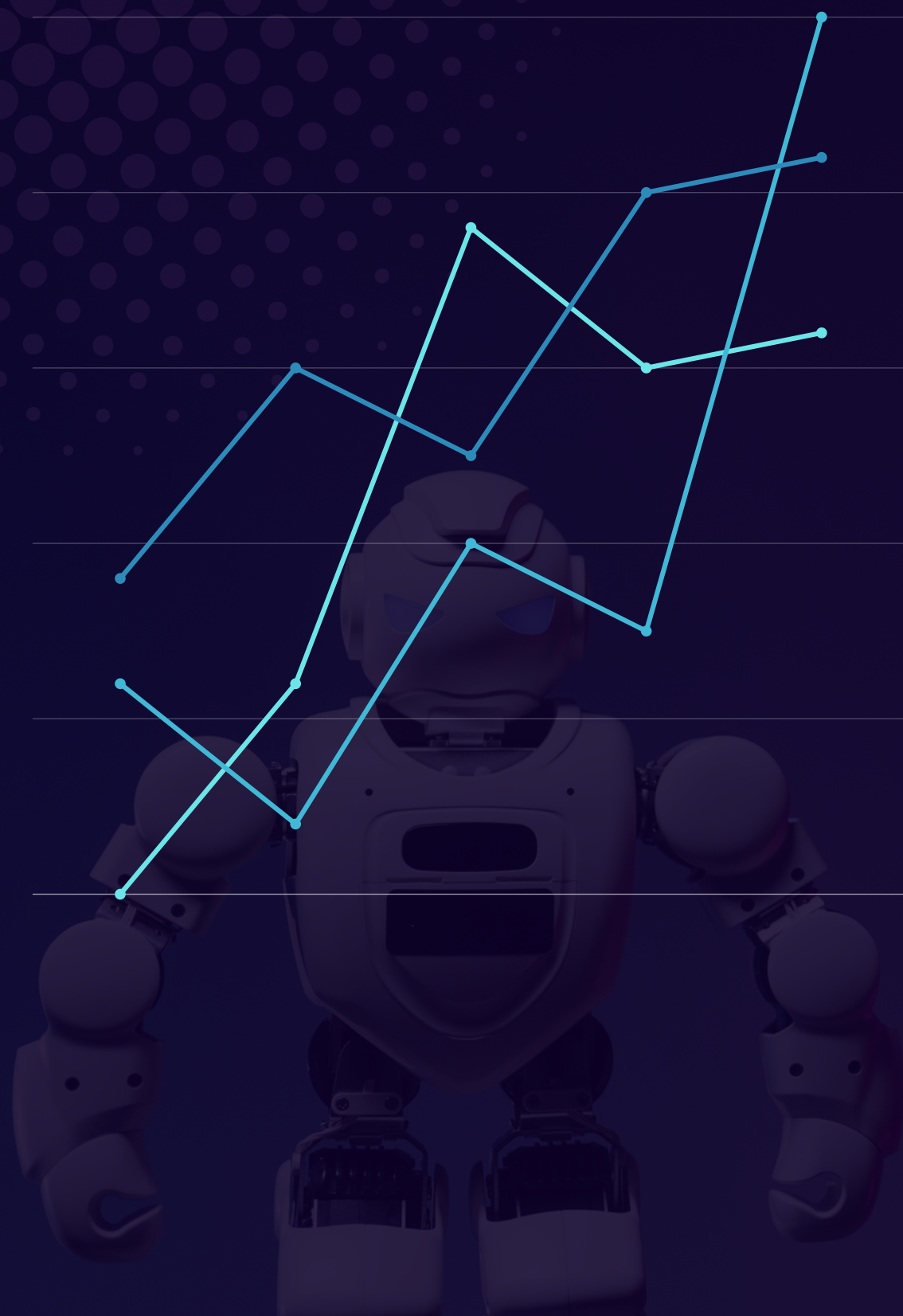
Decision Tree

Interpretable, Non-
Linear, Hierarchical and
probabilistic



Evaluation Metrics

We define appropriate metrics to assess the performance of our AI models, such as accuracy, precision, recall, F1 score and Confusion Matrices. These metrics help us quantify the effectiveness of our solutions.





Business Idea & Value Proposition

- **Product:** AI-based heart disease prediction system
- **Value:** Provides fast, accurate, and cost-effective heart disease risk assessment
- **For Hospitals & Clinics** – Helps doctors make quicker decisions.
- **For Insurance Companies** – Helps assess customer health risks.
- **For Individuals** – Enables early detection and lifestyle changes.



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Future Directions

- **Deep Learning Models** – Use Neural Networks (CNN, LSTM) for improved predictions.
- **Feature Engineering** – Extract more meaningful features from patient data.
- **Real-Time Prediction System** – Deploy as a web or mobile application for real-world use.
- **Larger & Diverse Dataset** – Improve model generalization with more medical records.



Thank You!